

FruitLearn AI: 30 Technical Viva Voce Questions & Answers

Comprehensive preparation guide covering SSL, ViTs, MAE, DINOv2, iBOT, ResNet-18, metrics, and dataset methodology.

Q1: What is Self-Supervised Learning (SSL)?

Self-supervised learning is a machine learning paradigm where models learn visual feature representations from unlabeled images by solving auxiliary pretext tasks (e.g. predicting masked patches or aligning augmented views) without relying on manual targets.

Q2: Why choose SSL for agricultural fruit quality analysis?

Manual fruit annotation is expensive, subjective, and slow. SSL allows models to leverage large unannotated agricultural image datasets to learn surface decay, texture, and shape features automatically.

Q3: How does Stage 1 pretraining differ from Stage 2 fine-tuning?

Stage 1 pretraining uses unlabeled images to optimize a pretext task (e.g. self-distillation). Stage 2 attaches a linear classification head and fine-tunes parameters on labeled samples.

Q4: What is the main mechanism behind DINOv2?

DINOv2 uses Student and EMA Teacher networks with multi-crop augmentations. The student learns to match teacher projections using centering and sharpening cross-entropy distillation loss without negative pairs.

Q5: What is the difference between MAE and DINOv2?

MAE reconstructs masked raw pixels using MSE loss, whereas DINOv2 aligns global feature vector distributions between augmented views using self-distillation.

Q6: What is the purpose of the ResNet-18 baseline in your project?

ResNet-18 is a traditional supervised CNN included to establish a baseline for evaluating how much self-supervised pretraining improves sample efficiency and accuracy.

Q7: How did you prevent data leakage?

Data leakage was prevented by executing a stratified 70/15/15 train/val/test split upfront with fixed seed 42, persisting split paths to `split_info.json`, and strictly reserving test images for final evaluation.

Q8: What happens when labeled training data is reduced to 10%?

The supervised ResNet-18 baseline drops to 60.00% accuracy, whereas DINOv2 retains 81.67% accuracy, proving that self-supervised pretrained features drastically reduce manual label dependence.

Q9: Which model achieved the highest accuracy?

iBOT achieved 100.00% test accuracy and DINOv2 achieved 99.94% test accuracy, outperforming ResNet-18 (99.56%) and MAE (81.11%).

Q10: What is centering and sharpening in DINOv2?

Centering adds a running average vector to teacher outputs to prevent one dimension from dominating (collapsing), while sharpening divides outputs by temperature τ to encourage sharp, confident probability distributions.